Latika Liladhar Dekate

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SUMMARY

Data Science graduate student with research in sim-to-real reinforcement learning and projects in retrieval-augmented generation, computer vision and data pipelines. Builds ML systems and tests whether their results hold up.

EDUCATION

Arizona State University — Tempe, AZ
M.S., Data Science, Analytics and Engineering (Electrical Engineering); GPA: 3.89/4.00 — Expected Dec 2026

S.I.E.S. Graduate School of Technology — Mumbai, India
B.E., Electronics & Telecommunication Engineering; GPA: 3.71/4.00 — Aug 2020 – Jun 2024

TECHNICAL SKILLS

Languages & Data: Python, SQL, PostgreSQL, pandas, NumPy, PySpark, Neo4j, Kafka, Java, Bash
Machine Learning: PyTorch, scikit-learn, TensorFlow, Vision Transformers, GNNs, Model Evaluation, Error Analysis
LLMs & NLP: LangChain, RAG, GraphRAG, ChromaDB, Hugging Face, FinBERT, SpaCy, LLM Evaluation
Vision & Research: OpenCV, YOLO, RAFT, ORB-SLAM3, Reinforcement Learning, SUMO, CityFlow, SLURM
Engineering & Tools: Docker, Kubernetes, Go, Git, Linux, REST APIs, Streamlit, D3.js, Tableau, Claude Code, Codex

EXPERIENCE

Arizona State University, DaRL Lab, SCAI — Tempe, AZ
Research Intern, Sim-to-Real Traffic Signal Control Benchmark — May 2026 – Aug 2026
- Ran and validated action-delay experiments on ASU's Sol cluster with SLURM, troubleshooting failed runs.
- Trained RL agents in LibSignal with SUMO and CityFlow, and traced code paths across agents and environments.
- Co-authored Sim2Signal (3rd of 9 authors; arXiv:2609.01676, preprint under review at AAAI 2027).

Google Developer Student Clubs, SIES GST — Mumbai, India
ML & DS Coordinator — Jul 2022 – Apr 2024
- Co-led a 10-week ML and data science program for 100+ students, turning concepts into hands-on workshops.
- Mentored students through Python, data visualization, model development and project debugging.
- Organized hackathons and coding events where students applied concepts in collaborative projects.

Help For Coders — Mumbai, India
SEO Intern — Dec 2021 – Apr 2023
- Analyzed keyword, traffic, ranking and engagement data in Google Analytics, Search Console, Semrush and Excel.
- Translated search findings into title, description and heading changes, and monitored pages after the site adopted some.

PROJECTS

Stealthy Data Poisoning in RAG Systems | *Llama 3.1, LangChain, ChromaDB* — Mar 2026 – Apr 2026
- Built the dashboard, t-SNE embedding analysis and plots for a five-person RAG poisoning study on Natural Questions.
- Found detectors averaged 0.527 AUC across 5 variants, as poisoned and clean documents overlap in embedding space.
- Wrote a three-category failure taxonomy that groups failures by what the embedding sees, not by detector.

Smart-Sighted SLAM | *ORB-SLAM3, OpenCV, RAFT, YOLO, PyTorch* — Mar 2026 – May 2026
- Suppressed moving objects before ORB feature extraction with a four-person team, instead of rebuilding the backend.
- Cut Sim(3) RMSE from 16.9 cm to 2.6 cm on one TUM sequence (monocular, 3 runs) with Farneback masking.
- Compared 4 masking strategies (PyrLK, Farneback, RAFT optical flow, YOLO masks) on accuracy and tracking time.

Breast Cancer Histopathology: ViT + GNN Fusion | *PyTorch, ViT, GNN* — Oct 2025 – Dec 2025
- Designed frozen-encoder ViT and GNN fusion experiments on BreaKHis 40x images in a three-person study.
- Trained only an MLP fusion head on 1,020 combined features to keep compute modest and limit overfitting.
- Benchmarked fusion against the ViT: 98.3% vs 98.2% fold-1 accuracy, a single-fold result with no clear gain.

Scalable Urban Transportation Analytics | *Python, Kafka, Kubernetes, Docker, Neo4j* — Fall 2025
- Migrated a Dockerized Neo4j batch workflow on NYC taxi data to Kafka streaming on Kubernetes (Minikube).
- Implemented Kafka Connect ingestion with idempotent Cypher MERGE to prevent duplicates on stream replays.
- Debugged Kafka listener and in-cluster networking, then ran ETL, PageRank and BFS on the streaming pipeline.

RECOGNITION

NASA Space Apps Challenge 2025: Phoenix team winner with AstraNode; advanced to Global Judging Round
Registered Design (India): No. 415110-001, 2024 | **Google Cloud Career Practitioners Certificate:** Jan 2023